Question Bank

Unit 3: Battery Technology

Basics & Electrochemistry

1. Define galvanic cell. How does it produce electrical energy?
2. What is electrode potential? Explain oxidation potential and reduction potential.
3. What is EMF of a cell? State the factors on which it depends.
4. Write the cell representation for a galvanic cell with a suitable example.
5. Differentiate between galvanic cell and electrolytic cell.

Introduction to Batteries

6. What is a battery? Why are batteries important in modern technology?
7. Classify batteries into primary, secondary, and reserve types with examples.
8. Differentiate between primary and secondary batteries with suitable examples.
9. What are reserve batteries? Give their significance and one example.
10. Discuss the advantages and limitations of primary batteries.

Battery Characteristics

11. Define and explain the following terms:

- Voltage
- Capacity
- Energy density
- Power density

12. Explain the significance of cycle life and shelf life in batteries.
13. Differentiate between energy efficiency and power density of a battery.
14. What are the factors that affect the performance of a battery?
15. Why is energy density considered an important parameter for portable devices?

Specific Batteries

16. Describe the construction and working of a Nickel–Cadmium (Ni–Cd) battery.
17. State the advantages and disadvantages of Ni–Cd batteries.
18. Describe the construction and working of a Lithium-ion battery.
19. What are the advantages of lithium-ion batteries over Ni–Cd batteries?
20. Discuss the applications of lithium-ion batteries in daily life.

Applications & EVs

21. Explain the principle of working of an electric vehicle.
22. Discuss the role of batteries in the development of electric vehicles.
23. Why are lithium-ion batteries preferred in electric vehicles?
24. Write a short note on the environmental impact of batteries.
25. What are the future prospects of battery technology?